

LEARNING INTENTION

I can estimate, construct and measure angles in degrees using a protractor.

SUCCESS CRITERIA

- ☐ I can name angle types (acute, right, obtuse, etc.)
- ☐ I can estimate an angle before measuring
- ☐ I can use a protractor accurately

STRATEGY BOX

Angle types: **Acute** (less than 90°), **Right** (exactly 90°), **Obtuse** (between 90° – 180°), **Straight** (exactly 180°), **Reflex** (more than 180°).

WORKED EXAMPLE

An angle measuring 45° is an **acute** angle, because it is less than 90° .

PART A — WARM UP

1. Name the angle type: 30°
2. Name the angle type: 90°
3. Name the angle type: 150°
4. Name the angle type: 200°

PART B — CORE SKILLS

5. Estimate the size of a slice of pizza cut into 8 equal pieces (full circle = 360°). What is each angle?
6. Use a protractor to measure 3 angles drawn by your teacher and record their sizes below.
7. Draw an angle of exactly 60° using your protractor.
8. Show your working for Question 5 below.

ANGLES HELPER

Line up the protractor's centre point on the angle's vertex (corner), with the baseline along one arm of the angle. Read where the other arm crosses the scale — make sure you read the correct scale (inner or outer)!

WORKING SPACE — PRACTICE ANGLES

PART C — CHALLENGE

9. Real-World Context: A skateboard ramp has an incline of 35° . Is this angle acute, right, or obtuse? Sketch what this might look like.

10. Reasoning: Two angles sit on a straight line and add up to 180° . If one angle is 110° , what is the other angle? Explain how you know.

REFLECTION

Today I learned...

I CAN CHECKLIST

- ☐ I can name different angle types
- ☐ I can estimate angle size
- ☐ I can measure angles with a protractor

TEACHER FEEDBACK

Strength: _____

Next Step: _____

PARENT TIP

Spot angles around the house — door hinges, clock hands, roof lines — and guess their size.

TEACHER TIP

Have students estimate before measuring — builds angle sense, not just protractor skills.

ANSWER KEY (FOR PARENT / TEACHER USE)

Q1: 30° = acute

Q2: 90° = right angle

Q3: 150° = obtuse

Q4: 200° = reflex

Q5: $360^\circ \div 8 = 45^\circ$ per slice

Q6: Teacher-supplied angles — check against protractor reading

Q7: Check student's drawn 60° angle for accuracy

Q8: Working shown as above

Q9: 35° is acute (less than 90°)

Q10: $180^\circ - 110^\circ = 70^\circ$; angles on a straight line always sum to 180°